

Chapter 5: Mathematics Logic

Class 21: Practical / Activities (20 Minutes)

Topic: Visual Puzzle Activity, Shape Matching Challenge, Odd One Out Game & Logical Thinking Worksheet

Activity 1: Visual Puzzle Activity (5 Minutes)

Objective

To improve students observation, visual reasoning, pattern recognition, and logical thinking through picture-based puzzles.

Materials Required

- Visual Puzzle Flashcards
- Shape Cards
- Whiteboard & Marker
- Activity Worksheet
- Pencil & Eraser

Instructions

1. The teacher displays different picture and shape puzzles.
2. Students carefully observe each puzzle.
3. Students identify the missing shape, picture, or pattern.
4. Students explain why they selected their answer.
5. The teacher discusses the correct answers and reasoning.

Visual Puzzle Examples

Puzzle 1 – Complete the Pattern



Answer: 

Puzzle 2 – Find the Missing Shape



Answer: 

Puzzle 3 – Picture Sequence



Answer: 

Puzzle 4 – Which Picture Comes Next?



Answer: 

Teacher Observation Checklist

- ☒ Observes carefully.
 - ☒ Identifies patterns correctly.
 - ☒ Gives logical explanations.
 - ☒ Completes activities independently.
 - ☒ Participates actively.
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Activity 2: Shape Matching Challenge (5 Minutes)

Objective

To help students identify, compare, and match different geometric shapes while improving visual discrimination skills.

Materials Required

- Shape Flashcards
- Shadow Shape Cards
- Shape Worksheet
- Whiteboard

Instructions

1. Divide students into small groups.
2. Give each group a set of shape cards.
3. Students match each shape with its correct shadow or identical shape.
4. The fastest team with all correct matches wins.
5. Discuss the properties of each shape.

Shape Matching Activity

Shape	Match
Circle	○
Square	□
Triangle	△
Rectangle	□
Star	★
Oval	○

Challenge Round

Students sort the shapes based on:

- Number of sides
- Curved shapes
- Straight-edged shapes
- Closed shapes

Teacher Observation Checklist

- ✓ Recognizes shapes correctly.
 - ✓ Matches accurately.
 - ✓ Works well in a team.
 - ✓ Identifies shape properties.
 - ✓ Completes within the given time.
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Activity 3: Odd One Out Game (5 Minutes)

Objective

To develop classification skills, logical reasoning, and critical thinking by identifying objects that do not belong to a group.

Materials Required

- Picture Cards
- Flashcards
- Whiteboard
- Activity Worksheet

Instructions

1. The teacher shows four pictures or words.
2. Students identify the item that is different.
3. Students explain the reason for their choice.
4. Each correct answer earns one point.
5. The teacher encourages discussion when multiple answers are suggested.

Odd One Out Examples

Example 1

Apple – Mango – Banana – Bus

Answer: Bus

Example 2

Circle – Square – Triangle – Pencil

Answer: Pencil

Example 3

Dog – Cat – Cow – Book

Answer: Book

Example 4

Car – Bus – Train – Orange

Answer: Orange

Example 5

Rose – Lotus – Sunflower – Chair

Answer: Chair

Teacher Observation Checklist

- ☒ Identifies the different object.
- ☒ Gives correct reasoning.
- ☒ Thinks logically.
- ☒ Answers confidently.

- ✓ Participates enthusiastically.
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Activity 4: Logical Thinking Worksheet (5 Minutes)

Objective

To strengthen students reasoning, observation, comparison, and analytical thinking skills through simple logic exercises.

Materials Required

- Logical Thinking Worksheet
- Pencil
- Whiteboard
- Teacher Answer Key

Instructions

1. Distribute the worksheet to all students.
2. Students solve each question independently.
3. Review the answers together as a class.
4. Encourage students to explain how they arrived at each answer.
5. Provide positive feedback and guidance.

Worksheet Activities

Exercise 1 – Complete the Pattern

1. 2, 4, 6, 8, _____
 2. ★ ○ ★ ○ ★ _____
 3. 10, 20, 30, _____
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Exercise 2 – Find the Odd One Out

1. Apple – Banana – Mango – Table
2. Cat – Dog – Cow – Bus
3. Circle – Triangle – Square – Pencil

Exercise 3 – Match the Shapes

Match:

- Circle → ○
 - Square → □
 - Triangle → △
 - Star → ★
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Exercise 4 – Simple Logical Questions

1. Which number is greater?

45 or 54

2. Which number comes before 100?
 3. Teacher is to School as Doctor is to _____.
 4. Bird is to Nest as Bee is to _____.
 5. Which direction is opposite to Left?
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Teacher Observation Checklist

- ☒ Solves logical questions correctly.
 - ☒ Uses observation skills.
 - ☒ Applies reasoning independently.
 - ☒ Completes worksheet neatly.
 - ☒ Demonstrates confidence.
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Practical Time Distribution

Activity	Duration
Visual Puzzle Activity	5 Minutes
Shape Matching Challenge	5 Minutes

Activity	Duration
Odd One Out Game	5 Minutes
Logical Thinking Worksheet	5 Minutes
Total Practical Time	20 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Identify and solve **visual puzzles** using observation and logical reasoning.
- Match different **geometric shapes** accurately based on their properties.
- Recognize the **odd one out** by comparing objects and explaining logical reasons.
- Apply **critical thinking and analytical skills** to solve simple reasoning problems.
- Improve **observation, concentration, classification, and decision-making abilities** through interactive classroom activities.
- Build confidence in solving logical puzzles independently and participating actively in group-based mathematical activities.